

POWERFACTORY

PowerFactory 2024

Technical Reference

DIgSILENT Battery Energy Storage System Template

DIgSILENT BESS FrequencyCtrl 10kV 30MVA

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 Introduction

This document describes the battery template *BatteryWithFrequencyControl_10kV_30MVA* that is delivered with *PowerFactory* 14.1. The template is a generic model for a battery energy storing system (BESS). It represents the grid side converter and the battery (modelled in DSL). The model represents one BESS with a rated apparent power of 30 MVA it is connected on 10 kV voltage level. The BESS model could be used for:

- Load Flow Studies
- Stability Studies (RMS)
- Transient Studies (EMT)
- Balanced and unbalanced simulations (however control is implemented for the positive sequence only)

For dynamic simulation is a variable step size possible. The minimum step size or the fix step size should be lower than 5 ms.

2 How to use this Template

PowerFactory contains pre-defined templates of dynamic models for different generic wind turbines, photovoltaic generators and a battery energy storing system. Using the templates is simple, just click on the *General Templates* button ( ) and select the desired template in the following window. Thereafter you could place the model in the single line diagram. For this you have to click one time in the single line diagram, this shows the model connected to the mouse pointer a second click creates a new model of the selected template in the single line diagram. Each dynamic model consists of some elements in the single line diagram (mostly a terminal with a generator) plus a composite model, which is located in the recording grid.

After you placed all models in the single line diagram you have to close the selection window with the templates, this terminates the template inserting.

Do not delete the terminal to which the generator is connected - this would destroy the measurement points of all measurement devices (located in the corresponding composite model).

You could delete a placed model directly by pressing the undo button ( ) or you could delete the model also manually. If you want to delete a model manually you have to mind that the model consists of visible elements in the single line diagram and also of a composite model which is only visible in the Data Manager [4].

3 Model Description

The BESS is in the single line diagram represented by a static generator. The models of the controllers as well as the battery are collected in the composite models. The controller is located in the composite model *BESS-Control* and the battery model is located in the composite model *Battery*. This composite model for the control could be found either through the link *Plant Model* on the Basic Data page of the static generator or with the Data Manger in the used grid.

The composite model *BESS-Control* is created from the frame definition *Frame_BatteryCntrl* this frame is shown in Figure 3.1. The composite model *Battery* is created from the frame definition *Frame_Battery* this frame is shown in Figure 3.2.

The single slots are briefly described in Table 3.1 and Figure 3.2. Each slot could be filled with either a DSL-model, a measurement device or a *PowerFactory* element, such as a static generator. All measurement devices are connected either to the terminal or to the cubicle, which connects the static generator with the terminal. This approach ensures that all measurement devices are correctly connected after entering the model using the template.

Table 3.1: Frame description for *Frame_BatterCntrl* (*.BlkDef)

Slot Name	**Description**	**Needed Type**
Converter	Link to the static generator [1], representing the inverter	*.ElmGenstat
PQ-Control	Control of the active and reactive power of the inverter	DSL-Model
Frequency Control	Calculates a reference for the active power depending on the frequency	DSL-Model
PQ-Measurement	PQ-Measurement device (for feedback)	*.StaPqmea
Frequency Measurement	Frequency measurement for frequency control	*.ElmPhi
AC-Voltage	AC voltage measurement device for detecting faults and for voltage support	*.StaVmea
Charge Control	Calculates if the battery has to be charged or not	DSL-Model
Battery Model	Link to the composite model *Battery* [3]	*.ElmComp

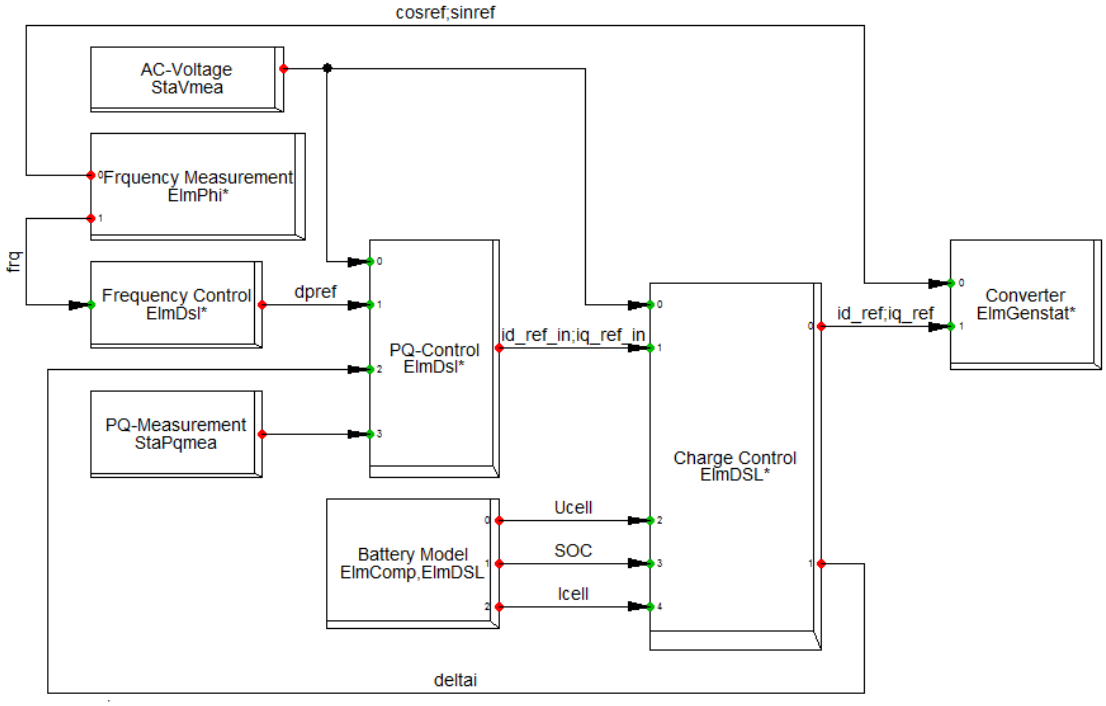


Figure 3.1: Frame Definition *Frame_BatterCntrl* (*.BlkDef)

Table 3.2: Frame description for *Frame_Battery* (*.BlkDef)

Slot Name	Description	Needed Type
Battery_Model	Model of the battery	DSL-Model
DC Side Calculation	Model of the DC-side (the static generator has no DC side)	DSL-Model
PQ Measurement (absolute Values)	PQ Measurement device which measures the active power on the AC side in absolute values	*.StaPqmea

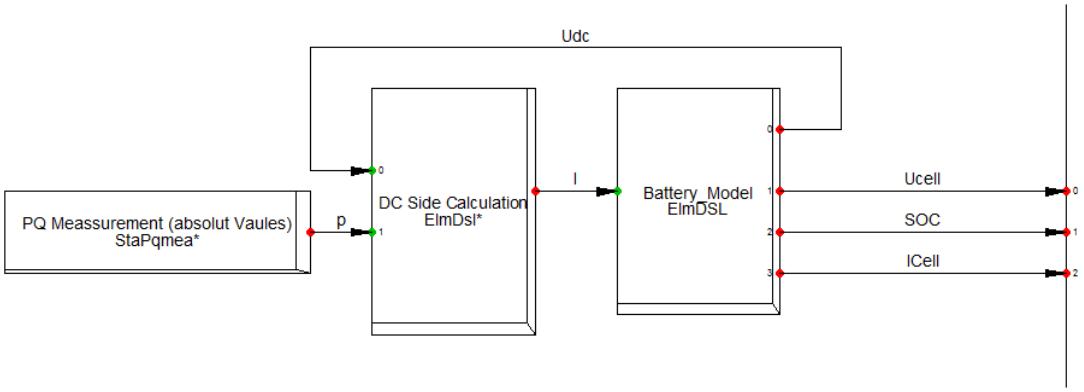


Figure 3.2: Frame Definition *Frame_Battery* (*.BlkDef)

4 How to Customise the Model

You will find in this chapter a description how you could configure the model from the template according to your specific needs. The configuration is sorted to the calculation functions because there are fewer configurations needed for load flow calculation than for a dynamic simulation.

The model is generic and therefore completely open - you could configure it according to your needs. Following you will find a short manual for changing some settings of the BESS. The BESS consists of a converter, which could operate as a rectifier (if the battery is charged) or as an inverter if the battery is discharged. There are therefore two main constraints on the system. The first constraint is the rated power/current of the converter and the second constraint is the capacity of the battery (i.e. the amount of stored energy).

For load flow calculations is only the rated power of interest because the battery capacity could only be used for a time domain simulation.

4.1 Load Flow

Changing the rated power

The rated power is for the load flow model only defined by the static generator data. If you want to change the rated power you have to change this on the basic data page and on the load flow page of the static generator.

Changing the nominal AC voltage

You could also change the nominal voltage of the converter if your BESS has a different nominal grid side voltage. The only modification you have to do is changing the nominal voltage of the terminal to which the static generator is connected.

Note: There is no saturation considered. It is assumed that the DC voltage is all the time high enough.

4.2 Short Circuit (VDE / IEC / ANSI / complete)

The short circuit behaviour of the model is also completely determined by the element settings. There are therefore no changes in the dynamic model parts necessary. It is important to note that the standard short circuit methods could not give precise results for a controlled BESS. For results that are more precise are dynamic simulations needed.

4.3 Dynamic Simulation (RMS/EMT)

The main purpose of the model created with the template is of course the dynamic simulation. The great advantage of using a template for creating the battery system model is that all measurement points are already correct defined and all common models are already configured.

The model is generic and therefore completely open - you could configure it according to your needs. Following you will find a short manual for changing some settings of the BESS. The BESS consists of a converter which could operate as a rectifier (if the battery is charged) or as an inverter if the battery is discharged. There are therefore two main constraints on the system. The first constraint is the rated power/current of the converter and the second constraint is the capacity of the battery (i.e. the amount of stored energy).

Changing the rated power

The rated power is for the load flow model only defined by the static generator data. If you want to change the rated power you have to change this on the basic data page and on the load flow page of the static generator.

Changing the nominal AC voltage

You could also change the nominal voltage of the converter if your BESS has a different nominal grid side voltage. The only modification you have to do is changing the nominal voltage of the terminal to which the static generator is connected.

Note: There is no saturation considered. It is assumed that the DC voltage is all the time high enough.

Disable reactive current support and voltage control

You could also disable the reactive current support and voltage control if the BESS in your project has not these features. The fastest way to do this is to remove or disable the AC voltage measurement:

1. Open the composite model *BESS-Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line

diagram. On the basic data page is a link to the composite model:



2. Select the *AC Voltage Measurement* and either disable the measurement or remove it from the composite model.

After these changes will the BESS generate all the time the reactive current given from the load flow initialisation.

Changing the reactive current support during faults

The BESS model contains also a reactive current support characteristic. This characteristic is activated during over- or undervoltage conditions (Figure 4.1). The parameter of the dead-band (AC_deadband) as well as the parameter for the slope (Kq) could be changed in the *PV Controller*.

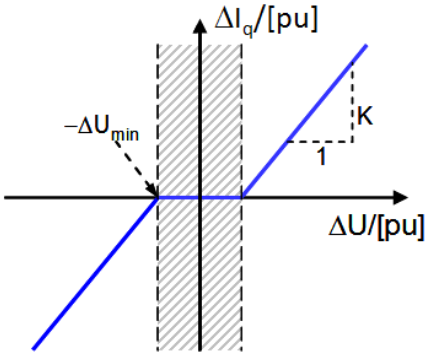


Figure 4.1: Reactive current support during voltage disturbances

1. Open the composite model *BESS-Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



2. Open in the composite model the *PV Controller* and make the changes according to your needs.

Limiting the charging or discharging current

Depending on the battery type the charging discharging current could be different. To consider this you have to change the limits in the common model *PV Controller*. The charging current is limited by the parameter *id_min* and the discharging current is limited by the parameter *id_max*. Both values are in per unit rated to the nominal power of the converter (static generator).

1. Open the composite model *BESS-Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



2. Open in the composite model the *PV Controller* and make the changes according to your needs.

Changing the initial state of charge of the battery

The initial state of charge (SOC) of the battery is stored in the battery model. To edit the battery models settings follow the instructions:

1. Open the composite model *BESS-Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



2. Doubly click on the *Battery* - this will open another composite model (described in Table 3.2).
3. Open in the composite model *Battery* the common model *Simple Battery*
4. Change the parameter *SOC0* according to your needs.

Changing the capacity of the battery

The total capacity is a product of the capacity of the single cell with the number of parallel cells. This information is stored in the battery model. To edit the battery models settings follow the instructions:

1. Open the composite model *BESS-Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



2. Doubly click on the *Battery* - this will open another composite model (described in Table 3.2).
3. Open in the composite model *Battery* the common model *Simple Battery*
4. Change the parameter *CellCapacity* and *CellsParallel* according to your needs.

Changing the charge control

The model contains also a control unit that prevents a discharging of the battery if the state of charge falls below a certain value (*minSOC*) and which prevents also a charging of the battery if the SOC is above a certain value (*maxSOC*). The charging control could also limit the active current in case of a voltage deviation to prefer the reactive current. The preference is changed if the AC-voltage deviates by a given value from the initial value (*deltaU*). For editing the charge control follow the steps below:

1. Open the composite model *BESS-Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



2. Doubly click on the *Charge Control*.
3. Change the parameters according to your needs.

5 References

[1] *Technical Reference "Static Generator"*. DIgSILENT GmbH, Gomaringen, Germany, 2019.

[2] DIgSILENT GmbH, Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany. *PowerFactory Manual. DIgSILENT PowerFactory Version 14.*, 2009.

[3] DIgSILENT GmbH, Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany. *Application Manual Battery Energy Storing System*, 2010.

[4] DIgSILENT GmbH, Heinrich-Hertz-Strasse 9, 72810 Gomaringen / Germany. *PowerFactory Manual. DIgSILENT PowerFactory Version 14.1*, 2011.

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